

Le Xuan Hoang

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Summary. AI researcher pursuing an M.S. in AI & Robotics at Sejong University, working at the Vision Language Intelligence Lab on computer-use agents. Background in Mechatronics Engineering with experience spanning vision-language models, computer vision, time-series forecasting, and business analytics. Built practical systems for facial-recognition attendance, electricity-demand forecasting, restaurant traffic prediction, and customer-behavior modeling.



Professional Experience

Research Assistant, VLILAB, Sejong University, Korea

2024.09 – Present

In Prep. **A Dataset for Computer-Use Agents** (1st author). Target: AAAI 2027.

Under Review. **DART: Density-Aware Adaptive Refinement for GUI Grounding on Computer-Use Agents** (1st author). IEEE Access.

Under Review. **Training-Free ROI Selection via Semantic Vector Fields for GUI Grounding** (2nd author). ECCV 2026.

Prior Submission. **CGAR: Confidence-Guided Adaptive Refinement for GUI Grounding** (1st author). Rejected from CVPR 2026.

- **Developed** a training-free adaptive refinement method for GUI grounding that leverages attention maps extracted from intermediate VLM layers as a signal to guide when to stop and how to refine. Evaluated on 8B-parameter models, improving ScreenSpot-Pro accuracy from 73.0% to 80.9%, achieving top-1 leaderboard performance, remaining competitive with much larger models, and surpassing existing refinement strategies.

- **Implemented** a VLM-based CCTV classification module using frame sampling and prompt inference on Qwen3-VL and InternVL3. Achieved 94.44% accuracy on the small test set and 83.53% accuracy on the larger CCTV-only dataset.

- **Developed** a vision-guided beamforming pipeline using YOLOv11 for detection and ByteTrack for vehicle tracking and position estimation to enable beam alignment. Evaluated on the DeepSense 6G dataset across three driving scenarios, improving average data rate by 20–29% over baseline.

Data Scientist, Center of Applied Data Science, FPT Corporation, Vietnam

2023.04 – 2024.08

- **Designed** monthly electricity-demand forecasting models using STL to decompose time series into trend, seasonal, and residual components, then applying separate linear, statistical parameter-based, and ARIMA-based models to each component. Achieved MAPE < 6% across all 13 provinces and < 4% in six, directly supporting energy planning.

- **Improved** a daily guest-count forecasting pipeline for 440+ restaurants by segmenting venues into stable and volatile traffic groups with separate XGBoost models. Added feature engineering, anomaly detection, missing-data imputation, and parameter tuning, reducing feature count by ~90% and training time by ~70%. Lowered WAPE from 25.3% to 22.0%, with 1.0–4.2% improvement across deployment months.

- **Processed** 96M+ raw web logs into 20 labeled user actions, mapped 32.5M sessions into a 5-phase customer journey and 3 behavioral segments, and delivered anonymous personas to support business strategy. Applied Markov chain models to predict section-level exit intent, achieving a 67% F1 score.

- **Developed** automated data enrichment pipelines using Selenium and BeautifulSoup across LinkedIn, Google Maps, and Chotot, extracting 1M+ records and 200K+ unique contacts to support customer analytics and lead recommendation.

Data Scientist, VTCODE Company, Vietnam

2022.06 – 2023.03

- **Built and deployed** a real-time facial-recognition attendance system using SSD detection, centroid tracking, and VGGFace2/ResNet50 embeddings with cosine similarity. Automated employee check-in/check-out in office environments, achieving 97% accuracy. Extended into a web platform, building the back-end with FastAPI REST APIs, MongoDB.

- **Built** daily restaurant revenue forecasting models using FBProphet to support financial planning.

- **Built** ETL pipelines integrating multi-source company data (databases, spreadsheets, documents) with crawled external web data to deliver analysis and recommendations to the board of directors.

Research Intern, Control and Automation Laboratory, HCMUT, Vietnam

2021.01 – 2022.06

- **Built** an automated chicken sorting system with a CNN classifier and multi-frame majority voting, achieving 95.9% accuracy at 4.56s per prediction (up from 91.5% single-frame). Designed the microcontroller-driven sorting mechanism with stepper motors and IR sensors for real-time operation.

Education

Master of Science, Sejong University, Seoul, Korea

2024 – 2026

Department: Artificial Intelligence and Robotics

Advisor: Associate Professor Cheol Jeong

Thesis: Confidence-Guided Adaptive Refinement Technique for GUI Grounding

GPA: 4.29/4.5

Bachelor of Engineering, VNU – Ho Chi Minh University of Technology, Vietnam

2017 – 2023

Department: Mechatronics Engineering (Excellent Engineers – PFIEV)

Advisor: Ph.D. Pham Cong Bang

Thesis: Analysis and Design of an Automated Chicken Classification System

GPA: 8.13/10

Additional Information

AI & ML. Vision-Language Models, GUI Grounding, Computer Vision, Object Detection, Face Recognition, Deep Learning, Time-Series Forecasting.

Programming. Python (PyTorch, scikit-learn, OpenCV, YOLO, FastAPI, Pandas, NumPy, PySpark, Selenium), C/C++.

Tools. Git, Linux, Jupyter Notebook, Office, MongoDB, Google Colab, SolidWorks, AutoCAD, MATLAB/Simulink.

Languages. Vietnamese (Native), English (Advanced), Korean (Basic).

Certifications. Machine Learning, Data Science, Deep Learning, STIC, Scientific Computing with Python, Data Analysis.